

Mazen Wael Mahmoud

Software Engineer | Backend & Full-Stack | AI/ML

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PROFESSIONAL SUMMARY

Software Engineer specializing in software development, web development, backend architecture, full-stack systems, and AI/ML pipelines. Demonstrated track record in building production RAG legal assistants with sub-5s response latency and architecting RESTful microservices that achieved 35% query latency reduction and 45% faster CI/CD deployments. Adept in Agile collaboration across FastAPI, ASP.NET Core, Docker, and SQL Server.

EDUCATION

B.Sc. Software Engineering — Computer Science

Cairo University, Cairo, Egypt

2022 – Jun 2026

Grade: Very Good | GPA: 3.02

TECHNICAL SKILLS

Programming Languages: Python, C#, C++, Java, JavaScript (ES6+), TypeScript, SQL

Backend & Frameworks: ASP.NET Core, FastAPI, Flask, .NET MVC, Entity Framework Core, RESTful API Design, WebSocket

Databases: SQL Server, MySQL, Database Schema Design, B-Tree Indexing, Query Optimization

DevOps & Tools: Docker, Linux (Ubuntu), Nginx, Git, GitHub Actions (CI/CD), Postman

AI & ML: Retrieval-Augmented Generation (RAG), LangChain, FAISS, Gemini API, Vector Embeddings, scikit-learn, TensorFlow, Keras, Pandas, NLP

Computer Vision: SAM 2.1, OpenCV, PyTorch, Hugging Face Transformers

EXPERIENCE

Software Engineering Intern

ALX Africa, Cairo, Egypt

Sep 2023 – Nov 2024

- Engineered scalable backend architectures, modular RESTful services, and automated CI/CD workflows, driving end-to-end delivery of full-stack production capstone systems.
- Collaborated within a 3-engineer Agile squad—including peer code review—to architect and deploy Market Mate; built 15+ Flask/MySQL RESTful endpoints and reduced database query latency by 35% via systematic B-tree indexing and N+1 query elimination.
- Automated end-to-end CI/CD pipelines via GitHub Actions and containerized Linux microservices with Docker and Nginx, cutting deployment cycle overhead by 45%.

PROJECTS

Arabic Hotel Reviews Sentiment Classification — Deep Learning NLP Competition [Github](#) Python · TensorFlow · Keras · Scikit-Learn · NLP

- Built an Arabic NLP preprocessing pipeline (diacritics/tatweel removal, letter normalization, negation preservation) and benchmarked five architectures—RNN, BiLSTM, BiGRU, Attention, and a custom Transformer—on 105K balanced reviews, raising accuracy from 50.76% to 94.87%.
- Built a 4-way ensemble blending BiGRU, BiLSTM, Attention, and a metadata-enriched BiGRU, with error analysis on mixed-sentiment and noisy cases, reaching 95.061% accuracy on the competition test set.

Eyego — Real-Time Object Tracking with SAM 2.1 [Github](#) Python · SAM 2.1 · OpenCV · PyTorch · Transformers

- Built a real-time webcam/video object tracker using Meta's SAM 2.1 Tiny, producing pixel-level segmentation masks from user-selected boxes, optimized for GPU inference.
- Implemented re-identification using aspect-ratio, HSV histogram, and rotation-invariant matching to reject look-alikes and recover the target when it re-enters the frame.

Market Mate — Full-Stack Marketplace Platform [Github](#) Flask · MySQL · Docker · GitHub Actions

- Led full-stack development on a marketplace platform with JWT authentication, a 50+ SKU product catalog, and real-time notifications via WebSocket, deployed with Nginx and Docker.
- Reduced API response latency by 30% through MySQL indexing and N+1 elimination; cut deployment overhead by 60% with a CI/CD pipeline.

Egyptian Legal AI — RAG-based Legal Assistant (Graduation Project) [GitHub](#) Python · FastAPI · LangChain · Gemini · React

- Architected a RAG system for Egyptian Labor Law using LangChain and FAISS, delivering accurate Arabic legal responses with cited article references.
- Built a production FastAPI backend with SQL Server chat history and JWT auth; wrote technical documentation and handled concurrent sessions with sub-5s latency and a 100% pass rate across 75 test cases.

Airbnb Clone — Distributed Booking API [GitHub](#) Flask · MySQL · HTML/CSS

- Designed and built a Flask RESTful API for booking and listing workflows across 5 normalized models, with full CRUD coverage and schema-level data integrity.
- Implemented a responsive HTML/CSS frontend consuming the API, covering the end-to-end booking and listing flow.

Heart Disease Prediction — ML Classification Model [GitHub](#) Python · scikit-learn · Matplotlib · Seaborn

- Trained logistic regression and random forest classifiers on the UCI Heart Disease dataset, achieving 87% accuracy after tuning with GridSearchCV.
- Conducted EDA across 13 clinical features and produced 8 visualizations surfacing feature correlations and class imbalances.

Habit Tracker App — ASP.NET Core CRUD Application [GitHub](#) C# · ASP.NET Core · SQL Server · EF Core

- Designed relational models with EF Core and built full CRUD controllers with Razor views in cross-platform ASP.NET Core MVC, delivering a production-style productivity tool backed by SQL Server.

CERTIFICATIONS

<u>Intermediate Machine Learning (Deep Learning)</u> <i>NetPoint</i>	Aug 2026 – Sep 2026
<u>Full Stack Software Engineering</u> <i>ALX Africa</i>	Sep 2023 – Nov 2024
<u>Deep Learning Specialization</u> <i>DeepLearning.AI (Coursera)</i>	Jun 2025 – Oct 2025
<u>AI & Machine Learning Program</u> <i>Microsoft / Sprints</i>	Jun 2025 – Nov 2025
<u>EF SET English Certificate — C2 Proficient (71/100)</u> <i>EF SET</i>	Mar 2025